

PROBLEM STATEMENT 5:

ARE INTERNSHIPS MORE VALUABLE THAN CERTIFICATIONS?



A Data-Driven Analysis to Unlock the Path to Better Careers and Higher Impact.



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INTERNSHIPS



CERTIFICATIONS



KEY AREAS WE ANALYZE



Placement
Outcomes



Salary
Impact



Career
Conversion



Employer
Preference



Skills &
Experience



Learning Path
Effectiveness



**DATA TURNS QUESTIONS
INTO CAREER CLARITY.**
Insights. Analysis. Impact.



This Project Helps ACC:

- ✓ Compare real impact of internships and certifications
- ✓ Identify the best learning pathways
- ✓ Improve placement outcomes
- ✓ Guide students with data-driven career recommendations

PROJECT OUTCOME

An Employability Intelligence & Career Success Analytics Platform that helps students make **smarter choices** and build **better futures**.



Better Learning
Pathways



Higher Placement
Success



AI-Powered Career
Recommendations



Stronger Future
Together

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Executive Summary

This report analyzes placement outcomes for 2,500 students graduating between 2023 and 2026, examining how college tier, degree program, certifications, internships, and demographics relate to placement rate, salary, and time-to-offer. The dataset shows an overall placement rate of 90.8% and an average placed salary of ₹13.16 LPA. Below are the headline findings; each is explored in full detail in the numbered sections that follow.

- **Experience beats pedigree and credentials alone:** Hybrid (internship + certification) candidates achieve 96.5–99.5% placement rates across every college tier, are offered roles 61% faster than certification-only candidates (39 vs. 101 days), and make up 90.3% of the top 10% of earners.
- **College tier and degree program matter less than the market assumes:** the Tier-1/Tier-3 placement gap nearly disappears once a student has real internship exposure, and salary spread across five degree types is under ₹0.5 LPA — far smaller than the certification-stacking or experience-profile effects.
- **Certifications and practical skills are statistically independent ($r = 0.08$):** they are earned through different activities and both drive outcomes largely on their own, which is the quantitative case for a genuinely hybrid strategy rather than doubling down on one path.
- **Direct Pre-Placement Offers have a hard duration floor near 5 months:** internships shorter than 5 months convert to a direct PPO 0% of the time in this dataset, while 7+ month internships convert 65.9% of the time and pay the highest average salary.
- **Skill-portfolio breadth is the steepest single predictor found in the analysis:** placement rate climbs from 57.5% to 99.1% and salary more than doubles as listed skills grow from 2 to 10.
- **No statistically significant gender pay gap was found ($p = 0.92$):** and the parity holds within every experience-profile band, not only in aggregate.
- **Internships function as a near-binary gate and certifications as a continuous multiplier:** on top of whatever internship tier a student has already reached — the two levers interact rather than simply adding together.

Institutional Tier Analysis

Grouping students by the six college categories in the dataset shows a persistent tier gradient in both placement rate and salary — but the gap narrows sharply once practical skill scores and experience profile are accounted for.

College Tier	Students	Placement Rate	Avg. Placed Salary (LPA)	Avg. Practical Score	Avg. Time to Offer (days)
IIT / NIT Group (Tier 1)	297	93.6%	₹15.56	73.5	52.2
BITS & Top National Inst (Tier 1)	320	95.9%	₹15.48	73.8	52.5
Apex Institute of Technology (Tier 2)	618	90.0%	₹13.19	68.4	51.5
State Tech University (Tier 2)	743	91.8%	₹12.75	67.0	56.0
City Engineering College (Tier 3)	314	85.7%	₹10.65	61.2	49.3
Regional Tech Institute (Tier 3)	208	85.1%	₹10.63	59.1	49.6

- **Tier premium is real but shrinking:** Tier-1 colleges post a ~10-point placement-rate lead and a ₹4.8–4.9 LPA salary premium over Tier-3 institutions.
- **Practical score tracks the same gradient:** average Practical_Skill_Score falls from ~73–74 at Tier 1 to ~59–61 at Tier 3, suggesting the salary gap is partly a skills-access gap rather than a pure brand effect.
- **A one-way ANOVA confirms the tier effect is statistically significant:** $F = 22.77$, $p < 0.001$ for practical skill score across the six college groups — the differences are not sampling noise.

Figure 1.1 — Placement Rate & Salary by College Tier

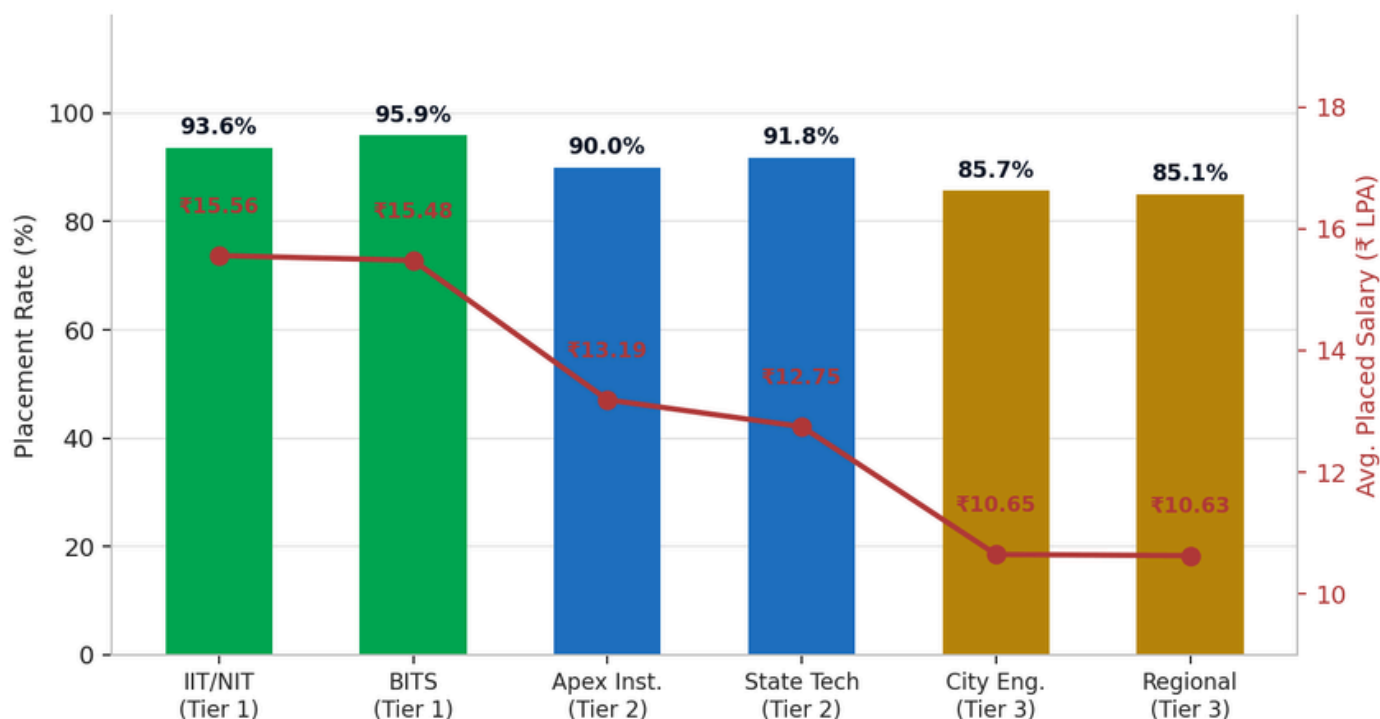


Figure 1.1 compares placement rates and average salaries across six colleges grouped into three tiers. Tier 1 colleges (IIT/NIT and BITS) perform best on both fronts, with placement rates above 93% and average salaries around ₹15.5–15.6 LPA. Tier 2 colleges (Apex Institute and State Tech) maintain strong placement rates of 90–92%, but their average salaries drop noticeably to the ₹11–13 LPA range. Tier 3 colleges (City Engineering and Regional) have the lowest placement rates, around 85%, and the lowest salaries, close to ₹10.6 LPA for both.

The bars (measured on the left axis) show placement percentage, while the red line with markers (measured on the right axis) tracks average placed salary in lakhs per annum. Overall, the chart illustrates a clear downward trend as college tier decreases — placement rates decline modestly, but salaries fall much more sharply, showing that the biggest disadvantage of lower-tier colleges lies in compensation rather than in getting placed at all.

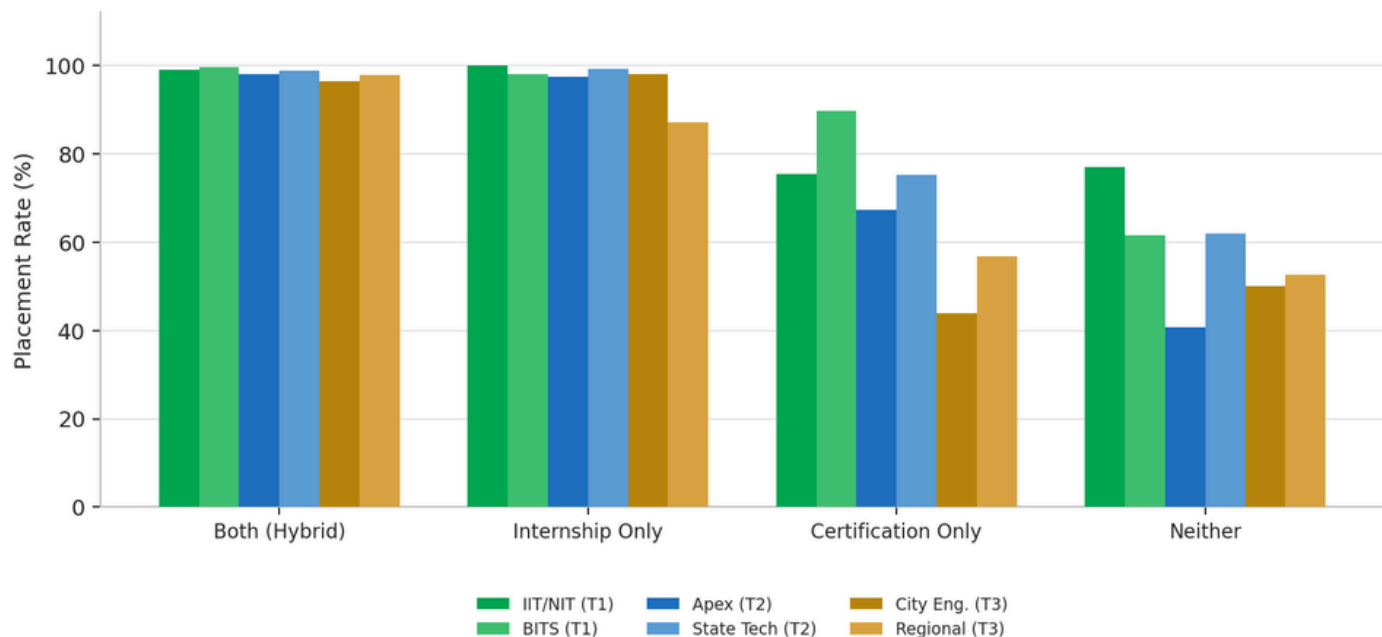
1.1 Does experience profile close the tier gap?

Cross-tabulating college tier against experience profile isolates whether a hybrid (internship + certification) strategy compensates for institutional tier.

College Tier	Both (Hybrid)	Internship Only	Certification Only	Neither
IIT / NIT Group (Tier 1)	98.9%	100.0%	75.4%	76.9%
BITS & Top National Inst (Tier 1)	99.5%	98.0%	89.7%	61.5%
Apex Institute of Technology (Tier 2)	98.0%	97.5%	67.3%	40.7%
State Tech University (Tier 2)	98.8%	99.1%	75.2%	61.9%
City Engineering College (Tier 3)	96.5%	97.9%	43.9%	50.0%
Regional Tech Institute (Tier 3)	97.7%	87.0%	56.8%	52.6%

This data shows how placement rates vary depending on the preparation pathway students take, broken down by college tier. The clearest pattern is that having both an internship and a certification or an internship alone produces near-universal placement (96–100%) across every tier, with almost no gap between Tier 1 and Tier 3 colleges. Certification alone is far weaker, especially at lower tiers — it drops to just 43.9% at City Engineering College, compared to 89.7% at BITS. Doing neither consistently yields the worst outcomes, falling as low as 40.7% at Apex Institute. The big takeaway: internship experience appears to be the dominant driver of placement success — colleges with weaker brand reputation (Tier 3) can still achieve near-Tier-1 placement rates if students secure internships, but they fall sharply behind when students rely on certifications or credentials alone.

Figure 1.2 — Placement Rate by College Tier × Experience Profile



The pattern is very clear: for "Both (Hybrid)" and "Internship Only," all six colleges cluster tightly near 97–100%, meaning tier barely matters when students have internship experience. But once you move to "Certification Only" and "Neither," the bars spread out dramatically — Tier 1 schools (green) still hold up reasonably well, while Tier 3 schools (gold/orange) and even some Tier 2 schools drop sharply, with City Engineering falling to around 44% for certification-only and 50% for neither. The key insight this visual reinforces: college tier prestige matters far less than practical experience. Internships act as a "great equalizer," while lack of experience widens the gap between elite and lower-tier institutions.

Key insight — the tier gap is largely a "Certification Only" and "Neither" problem

In the Both (Hybrid) column, placement rates cluster tightly between 96.5% and 99.5% regardless of tier — a spread of just 3 points. In the Certification Only column, the spread balloons to 45.8 points (43.9% to 89.7%). Institutional tier matters far less once a student has real internship exposure; it matters most for students relying on certifications alone, where Tier-1 brand recognition appears to substitute for the practical-skill signal a certificate alone doesn't provide.

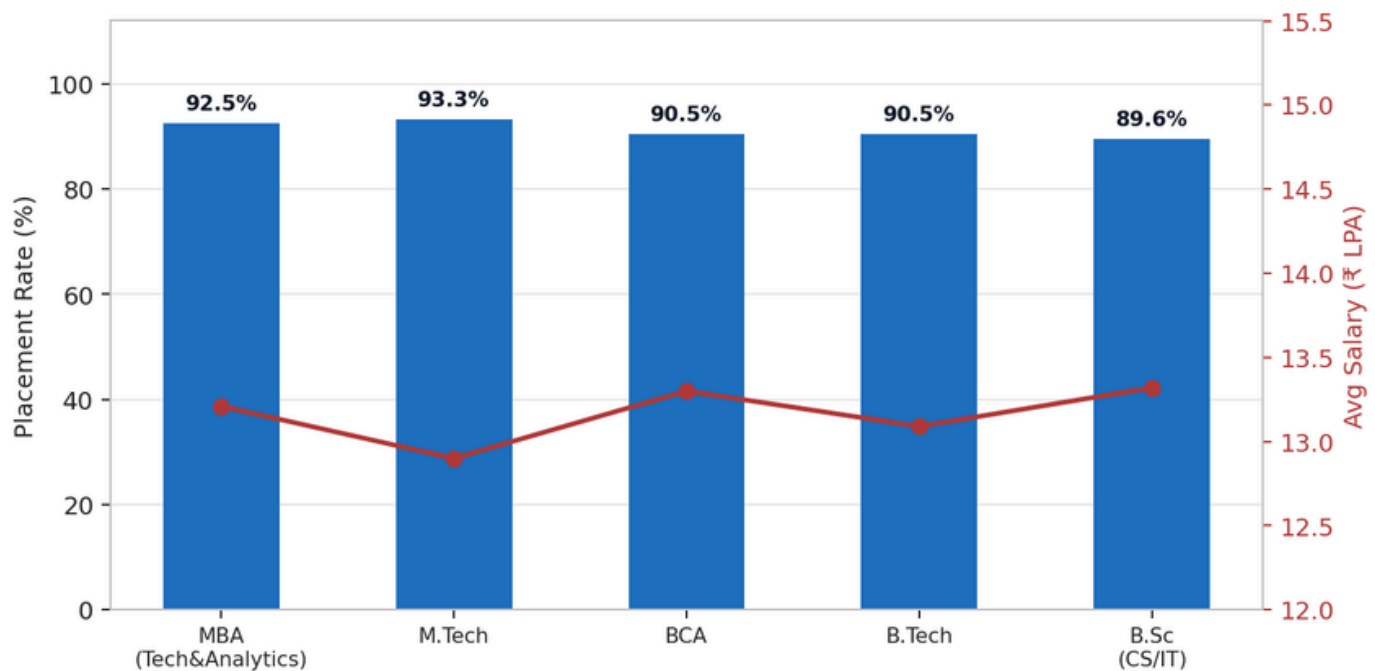
Degree Program Comparison

Comparing the five education levels shows outcomes are far more compressed across degree type than across college tier — a signal that programme choice matters less than what a student does during the programme.

Degree	Students	Placement Rate	Avg. Salary (LPA)	Avg. Theory Score	Avg. Time to Offer
MBA (Tech & Analytics)	186	92.5%	₹13.21	68.0	50.0 days
M.Tech	194	93.3%	₹12.90	67.3	54.7 days
BCA	493	90.5%	₹13.30	68.2	51.5 days
B.Tech	1,289	90.5%	₹13.09	67.5	53.0 days
B.Sc (CS/IT)	338	89.6%	₹13.32	67.4	53.6 days

The spread in average placed salary across all five degree types is under ₹0.5 LPA (₹12.90–₹13.32), and placement rates sit within a 4-point band (89.6%–93.3%). This is a materially smaller spread than either the college-tier gap or the certification-stacking effect — reinforcing that experience and skills variables outweigh the degree label itself.

Figure 2.1 — Placement Rate & Salary by Degree Program



This chart looks at placement rates and average salaries across five degree programs. Unlike the college-tier comparisons, the differences here are fairly narrow — placement rates range from 89.6% (B.Sc CS/IT) to 93.3% (M.Tech), all clustered within about a 4-point band.

Salaries, shown by the red line on the right axis, hover between roughly ₹13.0–13.3 LPA across all programs, with M.Tech actually showing the lowest average salary despite having the highest placement rate, and BCA and B.Sc (CS/IT) edging slightly higher on pay.

The overall takeaway is that degree program alone doesn't create the kind of dramatic spread seen with college tier or experience type — placement rates and salaries stay relatively consistent regardless of which program a student pursues, suggesting factors like internships and hands-on experience (from the earlier charts) matter more than the specific degree.

2.1 Branch-level detail

Branch	Students	Placement Rate	Avg. Salary (LPA)	Avg. Time to Offer
Computer Science & Engineering	884	90.8%	₹13.06	52.3 days
Data Science & Artificial Intelligence	640	91.2%	₹13.40	52.1 days
Information Technology	332	91.6%	₹12.92	54.3 days
Electronics & Communication	308	89.0%	₹12.87	55.9 days
Business Analytics	201	90.5%	₹13.35	50.0 days
Cyber Security	135	90.4%	₹13.55	50.7 days

Cyber Security holds the highest average salary (₹13.55 LPA) despite being the smallest branch by enrolment (135 students), while Electronics & Communication lags on every metric — the widest branch-level gap in the dataset outside of the tier and experience-profile splits.

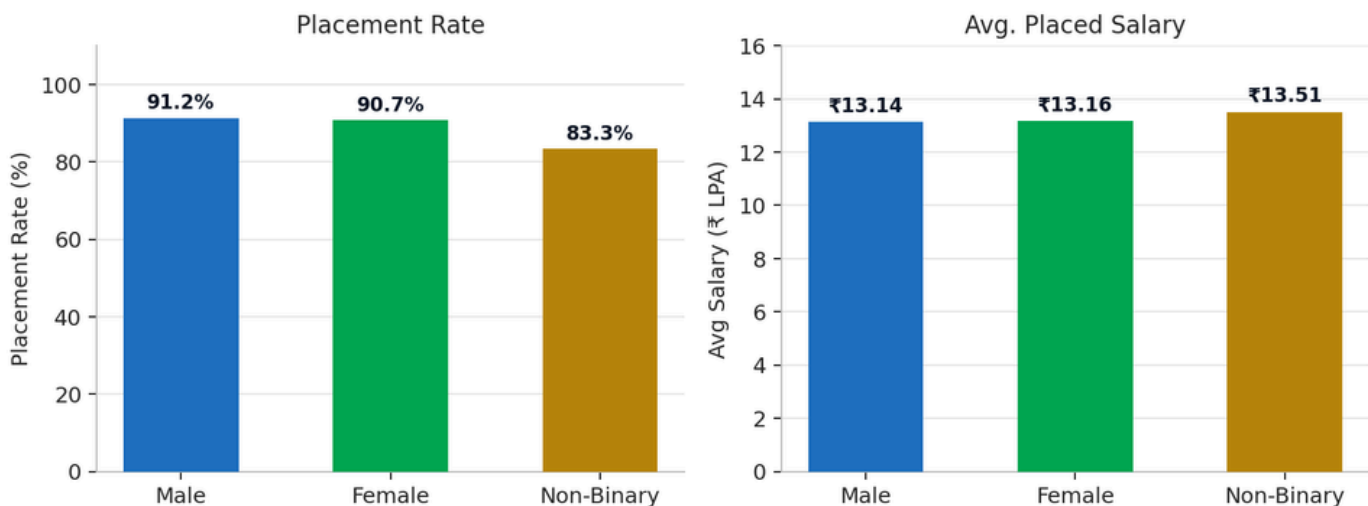
Demographic Equity Check: Gender & Age

3.1 Gender

Gender	Students	Placement Rate	Avg. Salary (LPA)	Avg. Employer Pref. Score
Male	1,318	91.2%	₹13.14	7.4 / 10
Female	1,116	90.7%	₹13.16	7.4 / 10
Non-Binary	66	83.3%	₹13.51	7.4 / 10

A Welch's t-test on placed salaries between male and female students returns $t = 0.10$, $p = 0.92$ — no statistically significant salary gap. A chi-square test of gender against placement outcome gives $p = 0.098$, i.e. not significant at the conventional 0.05 threshold, though the Non-Binary group's 83.3% placement rate ($n = 66$) is the one cell worth monitoring as the sample grows, since a smaller base makes the estimate less stable.

Figure 3.1 — Placement Rate & Salary by Gender



Intersectional check: gender × experience profile (average salary, LPA)

Gender	Both (Hybrid)	Internship Only	Certification Only	Neither
Female	₹14.07	₹13.07	₹6.24	₹3.98
Male	₹14.19	₹12.88	₹6.22	₹4.24
Non-Binary	₹13.78	₹11.77	₹4.12	₹0.00 (n<5)

Within every experience-profile band, male and female averages sit within a few thousand rupees of each other — the near-parity in the headline gender table holds up once experience profile is controlled for. The larger swings are between experience-profile columns (as much as ₹10 LPA), not between genders within a column, reinforcing that experience strategy — not gender — is the dominant salary lever in this dataset.

3.2 Age Band

Age Band	Students	Placement Rate	Avg. Salary (LPA)	Avg. Time to Offer
20–22	1,631	90.0%	₹13.15	52.8 days
23–24	738	92.3%	₹13.12	53.1 days
25–26	131	91.6%	₹13.42	48.9 days

Age shows negligible correlation with any outcome variable ($r \approx 0.01$ against salary and employer-preference score), confirming that experience quality, not age or repeated attempts, is what the market is pricing.

Graduation-Year Cohort Trends (2023–2026)

Tracking outcomes by graduation year reveals a steady, monotonic improvement — useful for distinguishing genuine skill-building trends from one-off cohort noise.

Grad. Year	Students	Placement Rate	Avg. Salary (LPA)	Avg. Practical Score	Avg. Time to Offer
2023	502	88.6%	₹13.18	66.9	53.2 days
2024	785	90.3%	₹13.08	66.9	53.4 days
2025	875	91.5%	₹13.06	67.4	53.7 days
2026	338	92.9%	₹13.54	70.9	47.7 days

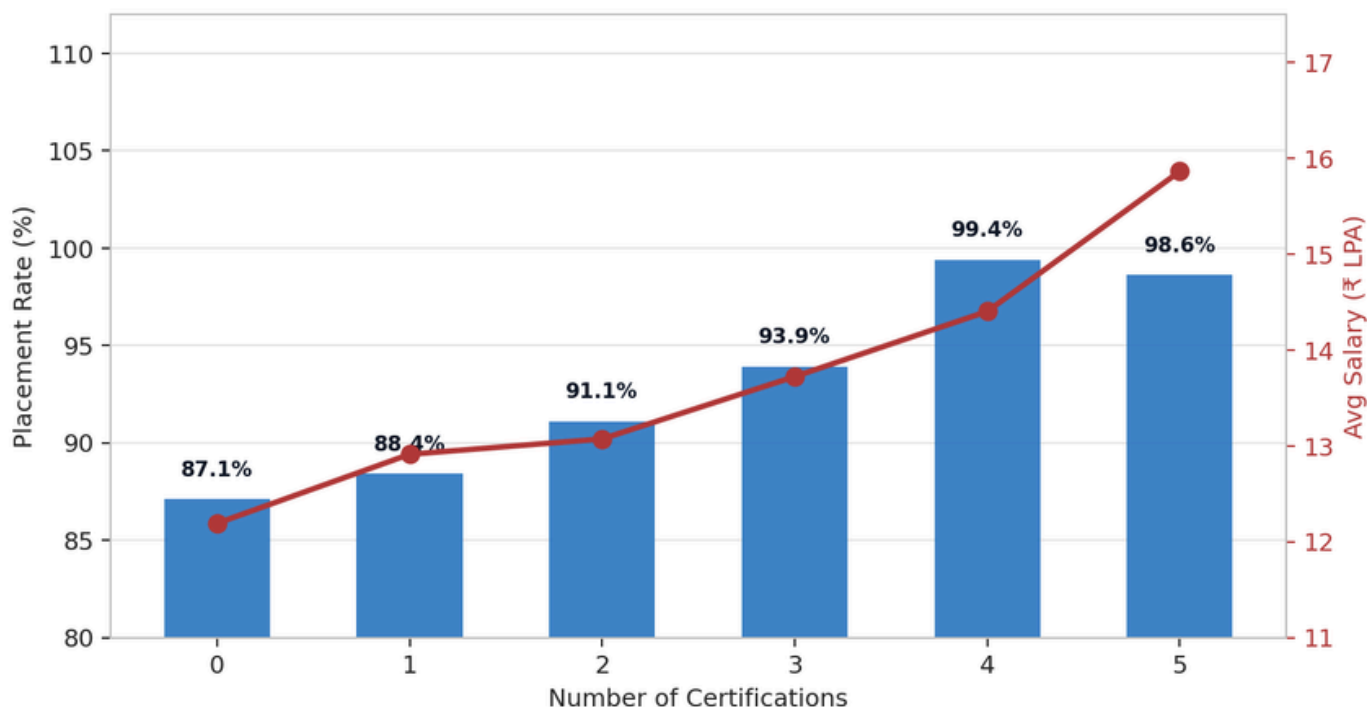
- **Placement rate has risen 4.3 points in four years** (88.6% → 92.9%), even as cohort size grew, indicating the improvement is not simply a smaller, more selective graduating class.
- **2026 is the inflection cohort:** practical skill score jumps from ~67 (2023–25 average) to 70.9, and time-to-offer compresses from ~53 days to 47.7 days — consistent with more recent graduates entering with stronger internship exposure baked into their profile from the start.
- **Salary held flat for three years before rising:** average placed salary barely moved between 2023 and 2025 (₹13.18 → ₹13.06), then jumped to ₹13.54 in 2026 — the salary gain trails the skill-score gain by roughly a year, consistent with employer compensation benchmarks adjusting with a lag behind visible talent-quality improvements.

Certification Stacking: Where Do Returns Diminish?

Rather than treating certifications as binary (have/don't have), grouping by the exact count held shows a clear — and clearly non-linear — dose-response curve.

# Certifications	Students	Placement Rate	Avg. Salary (LPA)	Avg. Theory Score	Avg. Time to Offer
0	479	87.1%	₹12.19	47.4	63.6 days
1	731	88.4%	₹12.91	59.0	55.5 days
2	661	91.1%	₹13.07	71.7	53.1 days
3	377	93.9%	₹13.72	84.1	47.6 days
4	181	99.4%	₹14.40	94.2	36.0 days
5	71	98.6%	₹15.86	99.2	27.1 days

- **The steepest jump is 0 → 1 certification:** placement rate barely moves (+1.3 pts) but this is the floor; the real acceleration starts at 2+ certifications, where placement rate crosses 91% and keeps climbing.
- **4 certifications is the practical sweet spot:** 99.4% placement and the largest single-step time-to-offer compression (47.6 → 36.0 days). Five certifications adds salary (₹15.86 LPA, the dataset's highest bracket) but placement rate ticks down slightly to 98.6%, and the sample thins to 71 students — a soft signal of diminishing marginal placement-probability returns beyond 4, even as pay for the credential-heavy segment keeps rising.
- **Certifications track theory, not practice:** the near-perfect climb in Theoretical_Score (47.4 → 99.2) alongside flat Practical_Skill_Score confirms certifications are a knowledge-signal, not a hands-on-skill signal — the two experience levers are complementary, not substitutes.

Figure 5.1 — Certification Stacking: Placement Rate & Salary

This chart examines "certification stacking" — how placement rate and salary change as students accumulate more certifications, from 0 up to 5.

There's a strong positive trend: placement rate climbs steadily from 87.1% with zero certifications to a peak of 99.4% at four certifications, while average salary rises even more sharply, from around ₹12.2 LPA at zero certifications to nearly ₹15.9 LPA at five. Notably, at five certifications, placement rate dips slightly to 98.6% (down from 99.4% at four), even though salary continues to climb — suggesting a possible point of diminishing or slightly negative returns on placement rate specifically, even as pay keeps increasing.

Overall, the chart makes a compelling case that stacking multiple certifications meaningfully boosts both employability and compensation, with the strongest gains occurring between 0–4 certifications, and salary benefits continuing even at the margin.

Certification Provider Effectiveness Ranking

Students often hold certifications from more than one provider (e.g. "Coursera; AWS"); each provider is credited separately below to compare like-for-like effectiveness across the 2,500-student pool.

Provider	Appearances	Placement Rate	Avg. Salary of Holders (LPA)
Udemy	463	95.5%	₹13.49
AWS	496	94.8%	₹13.51
NPTEL	526	93.5%	₹13.65
IBM	516	93.0%	₹13.64
Microsoft Azure	488	92.6%	₹13.72
Google Cloud	504	92.1%	₹13.47
Coursera	486	92.4%	₹13.61
Meta	461	89.6%	₹13.60

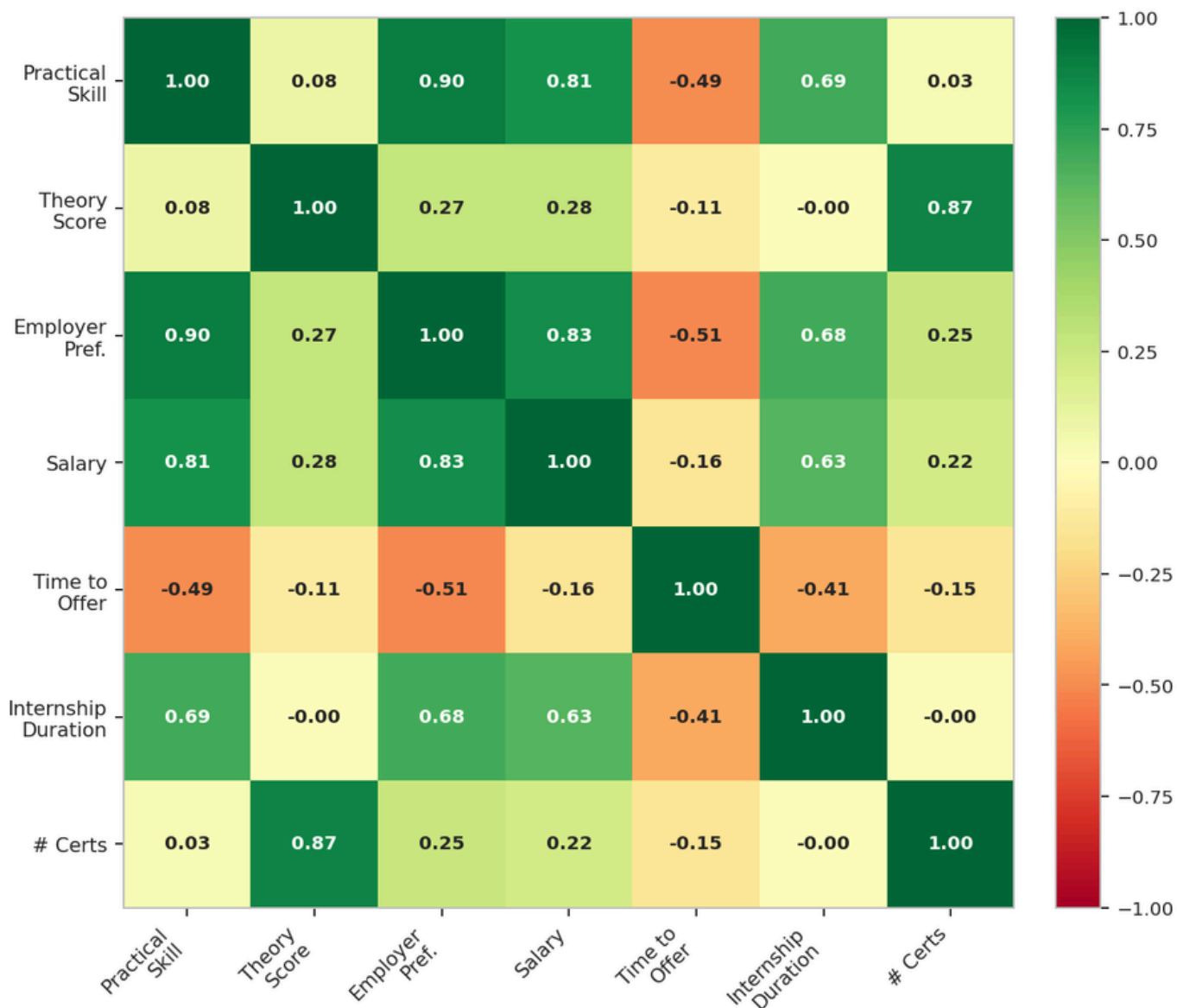
Ranked by average salary of holders, Microsoft Azure (₹13.72 LPA), NPTEL (₹13.65 LPA) and IBM (₹13.64 LPA) certifications are associated with the highest pay outcomes; ranked by placement rate, Udemy (95.5%) and AWS (94.8%) lead. Meta trails on both measures. Because students typically stack 1–3 providers together, these are association, not causal, rankings — the effect is best read as "which provider tends to appear in the strongest overall profiles," not "which single certificate to buy."

Statistical Drivers of Placement & Salary

7.1 Correlation matrix

Variable Pair	Correlation (r)	Read
Practical Skill Score ↔ Employer Preference Score	0.90	Very strong — practical skill is effectively what employers are scoring
Practical Skill Score ↔ Salary	0.81	Strong — hands-on skill converts directly to pay
Employer Preference Score ↔ Salary	0.83	Strong
Internship Duration ↔ Practical Skill Score	0.69	Strong — longer internships build the skill employers score
Number of Certifications ↔ Theoretical Score	0.87	Very strong — confirms Section 5's knowledge-signal finding
Practical Skill Score ↔ Theoretical Score	0.08	Near-zero — the two skill types are earned independently
Age ↔ any outcome variable	≤ 0.03	Negligible in every case

The headline finding is the near-zero correlation ($r = 0.08$) between practical and theoretical scores. They are not two measures of the same underlying ability — they are earned through different activities (internships vs. certifications/coursework) and both feed employer preference and salary largely independently. This is the statistical backbone for a genuinely "hybrid" strategy rather than a stronger version of either path alone.

Figure 7.1 — Correlation Matrix of Key Numeric Variables

This heatmap shows how seven key variables relate to one another, with values ranging from -1 (strong negative correlation, red) to +1 (strong positive correlation, dark green). The standout relationship is between Practical Skill and Employer Preference (0.90), along with Salary and Employer Preference (0.83) and Practical Skill and Salary (0.81) — showing that hands-on ability is closely tied to how employers rate candidates and what they pay. Interestingly, Theory Score correlates strongly with number of certifications (0.87) but has almost no relationship with practical skill (0.08) or employer preference (0.27), suggesting certifications may build theoretical knowledge more than practical competency. Time to Offer is negatively correlated with most positive outcomes, especially Employer Preference (-0.51) and Practical Skill (-0.49), meaning stronger candidates tend to get offers faster.

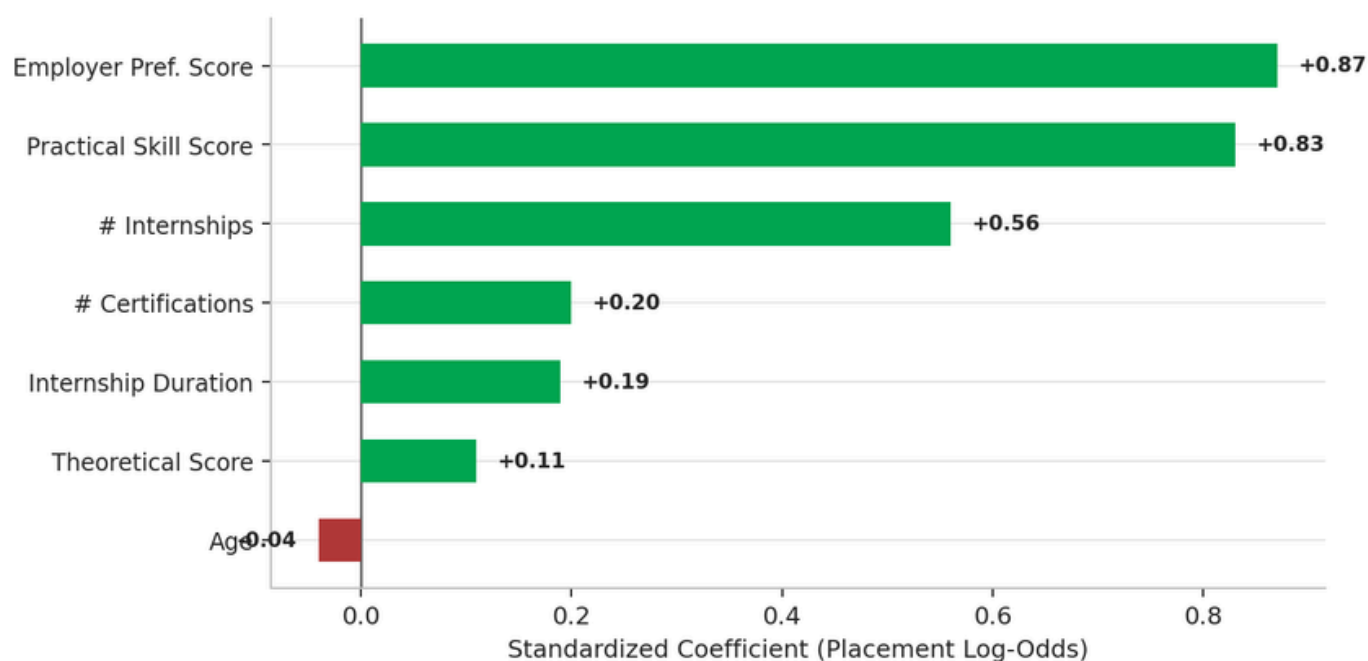
7.2 Logistic regression: standardized drivers of placement probability

A logistic regression on Placement_Status (Placed = 1) using seven standardized predictors reaches 91.2% training accuracy. Ranked by standardized coefficient (larger magnitude = larger swing in placement odds per one standard deviation of the input, holding others constant):

Rank	Predictor	Standardized Coefficient	Direction
1	Employer Preference Score	+0.87	↑ raises placement odds
2	Practical Skill Score	+0.83	↑ raises placement odds
3	Number of Internships	+0.56	↑ raises placement odds
4	Number of Certifications	+0.20	↑ raises placement odds
5	Internship Duration (months)	+0.19	↑ raises placement odds
6	Theoretical Score	+0.11	↑ raises placement odds
7	Age	-0.04	Effectively no effect

The ordering is instructive for advisory purposes: internship-adjacent variables (number of internships, duration) individually outweigh certification count in the placement model, even though certifications remain positive and additive — consistent with the "hybrid beats either alone" pattern the main dashboard already reports, but here isolated at the level of an individual predictive driver rather than a group average.

Figure 7.2 — Logistic Regression: Drivers of Placement



This chart presents results from a logistic regression model identifying which factors most strongly predict placement, using standardized coefficients (log-odds).

Employer Preference Score (+0.87) and Practical Skill Score (+0.83) emerge as by far the strongest positive predictors, confirming what the correlation matrix suggested — employers value practical ability above all else. Number of Internships (+0.56) is the third-strongest driver, reinforcing the "internships matter most" theme from earlier charts. Further down, Certifications (+0.20), Internship Duration (+0.19), and Theoretical Score (+0.11) all contribute positively but much more modestly. Age is the only negative factor (-0.04), though its effect is small.

Taken together with the earlier figures, this regression provides statistical confirmation of the overall narrative: practical, employer-facing competencies and real-world internship experience are the dominant forces behind placement success, while academic credentials like certifications and theory scores play a supporting — not leading — role.

7.3 Linear regression: standardized drivers of salary (placed students only)

A parallel ordinary-least-squares regression on Salary_Package_LPA for the 2,269 placed students, using the same seven standardized predictors, explains 78.9% of salary variance ($R^2 = 0.789$) — a strong fit for a single linear model on real-world outcome data.

Rank	Predictor	Standardized Coefficient	Direction
1	Theoretical Score	+1.44	↑ raises salary
2	Practical Skill Score	+1.34	↑ raises salary
3	Employer Preference Score	+1.00	↑ raises salary
4	Internship Duration (months)	+0.83	↑ raises salary
5	Number of Internships	+0.35	↑ raises salary
6	Age	−0.02	Effectively no effect
7	Number of Certifications	−0.69	Appears negative — see caveat below

Statistical caveat: multicollinearity, not a real penalty for certifications

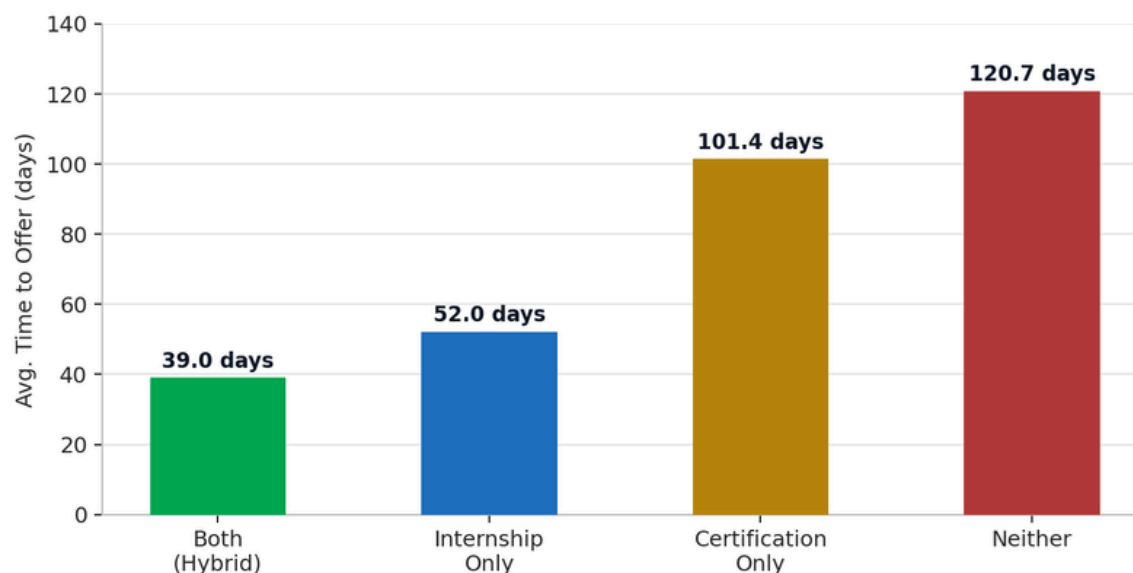
Number_of_Certifications shows a negative coefficient in this joint model despite a positive simple correlation with salary (Section 7.1, $r = 0.22$). This is a textbook multicollinearity artifact: because certifications correlate at $r = 0.87$ with Theoretical_Score, the model attributes their shared salary contribution almost entirely to Theoretical_Score once both are included together, leaving a small negative residual coefficient for certification count. This should not be read as "more certifications reduce salary" — Section 5's raw dose-response table (certifications climbing from ₹12.19 to ₹15.86 LPA) is the more reliable read on the certification-salary relationship. The two theoretical/practical-skill variables remain the dominant, non-redundant salary drivers in both models.

Time-to-Offer Analysis

Experience Profile	Mean Days	Median Days	Std. Dev.	Placed Students
Both (Hybrid)	39.0	40	25.4	1,519
Internship Only	52.0	54	27.2	347
Certification Only	101.4	103	17.4	333
Neither	120.7	120	14.0	70

Hybrid candidates are offered a role in roughly a third of the time of certification-only candidates (39.0 vs. 101.4 days — a 62-day / 61% reduction). A one-way ANOVA across the four experience-profile groups on placed salary is highly significant ($F = 376.6$, $p < 0.001$), confirming the profile effect is not sampling noise. Notably, the hybrid group also has the highest variance in time-to-offer ($\sigma = 25.4$), meaning outcomes are faster on average but less uniform — some hybrid candidates land offers almost immediately (fast-tracked via internship conversion) while others still go through a full recruiting cycle, whereas certification-only and "neither" candidates cluster more tightly around a slow, standard timeline ($\sigma = 17.4$ and 14.0).

Figure 8.1 — Avg. Time-to-Offer by Experience Profile



Internship Domain & Employer Analysis

Restricting to the 2,082 students who completed at least one internship, this section asks which internship domain and which host company are associated with the strongest downstream outcomes — an angle the dashboard's Skills Demand Engine does not cover at the domain/company level.

9.1 By internship domain

Domain	Interns	Placement Rate	Avg. Salary (LPA)	Direct PPO Rate	Avg. Duration (months)
Software Engineering	315	97.1%	₹14.39	23.2%	5.0
Product & Business Analytics	346	98.6%	₹14.34	22.8%	4.8
Artificial Intelligence / ML	288	99.7%	₹14.15	21.9%	4.8
Data Science & Analytics	306	98.7%	₹14.06	24.5%	4.9
Cloud & DevOps	308	98.4%	₹14.00	23.7%	4.7
Cyber Security	337	97.0%	₹13.96	19.9%	4.9

9.2 By host company (companies with ≥ 20 interns)

Company	Interns	Placement Rate	Avg. Salary (LPA)	Direct PPO Rate
Google India	196	100.0%	₹14.15	25.5%
Deloitte Consulting	184	98.4%	₹14.26	25.0%
FinTech Solutions	185	98.4%	₹14.18	21.6%
InnovateTech Startup	195	98.5%	₹14.13	20.5%
DataMetrics Labs	189	97.9%	₹14.29	22.8%
CyberGuard Analytics	188	97.9%	₹14.20	23.9%
Amazon AWS	191	97.9%	₹14.11	23.6%
Microsoft Tech	169	98.8%	₹14.02	20.1%
TCS Innovation Labs	205	97.6%	₹14.01	22.9%
Infosys Center	198	97.0%	₹14.17	20.2%

Google India is the only host company in the dataset with a 100% placement rate among its interns and also posts the highest direct-PPO rate (25.5%). Outcomes across all ten major host companies are tightly bunched (97.0%–100% placement, ₹14.01–14.29 LPA salary) — for students choosing between reputable hosts, domain fit and internship duration appear to matter more than the specific brand name once a company is already large enough to appear in this list.

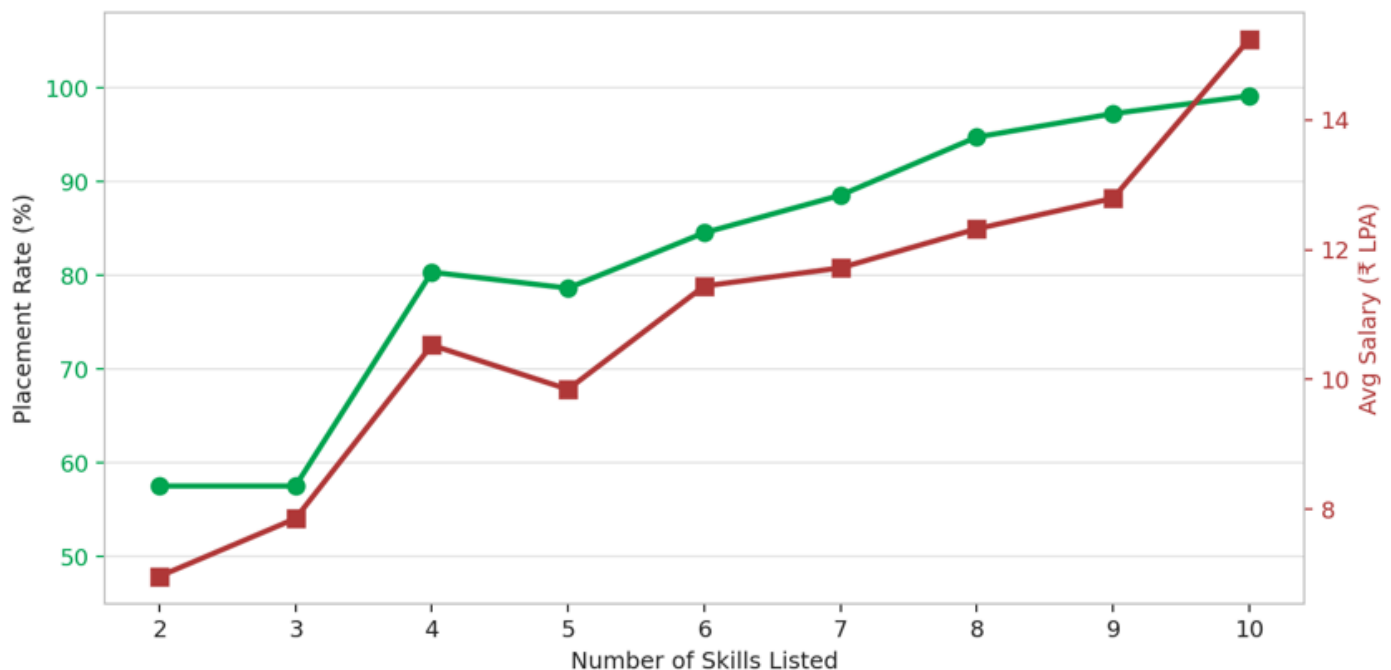
Skills Portfolio Analysis

The Primary_Skills field lists each student's self-reported skill tags (comma-separated). Exploding this field lets us look at both the effect of breadth (how many skills a student lists) and which specific individual skills co-occur with the strongest outcomes — a different lens from the dashboard's Skills Demand Engine, which addresses market demand rather than the returns to portfolio breadth.

Effect of skill-portfolio breadth

Students list 7.9 skills on average. Grouping by exact skill count shows one of the steepest gradients in the entire dataset:

# Skills Listed	Students	Placement Rate	Avg. Salary (LPA)
2	80	57.5%	₹6.97
3	106	57.5%	₹7.86
4	127	80.3%	₹10.53
5	187	78.6%	₹9.85
6	200	84.5%	₹11.44
7	218	88.5%	₹11.72
8	283	94.7%	₹12.32
9	218	97.2%	₹12.79
10	1,081	99.1%	₹15.24



This chart tracks how placement rate (green) and average salary (red) change as students list more skills in their portfolio, ranging from 2 to 10 skills.

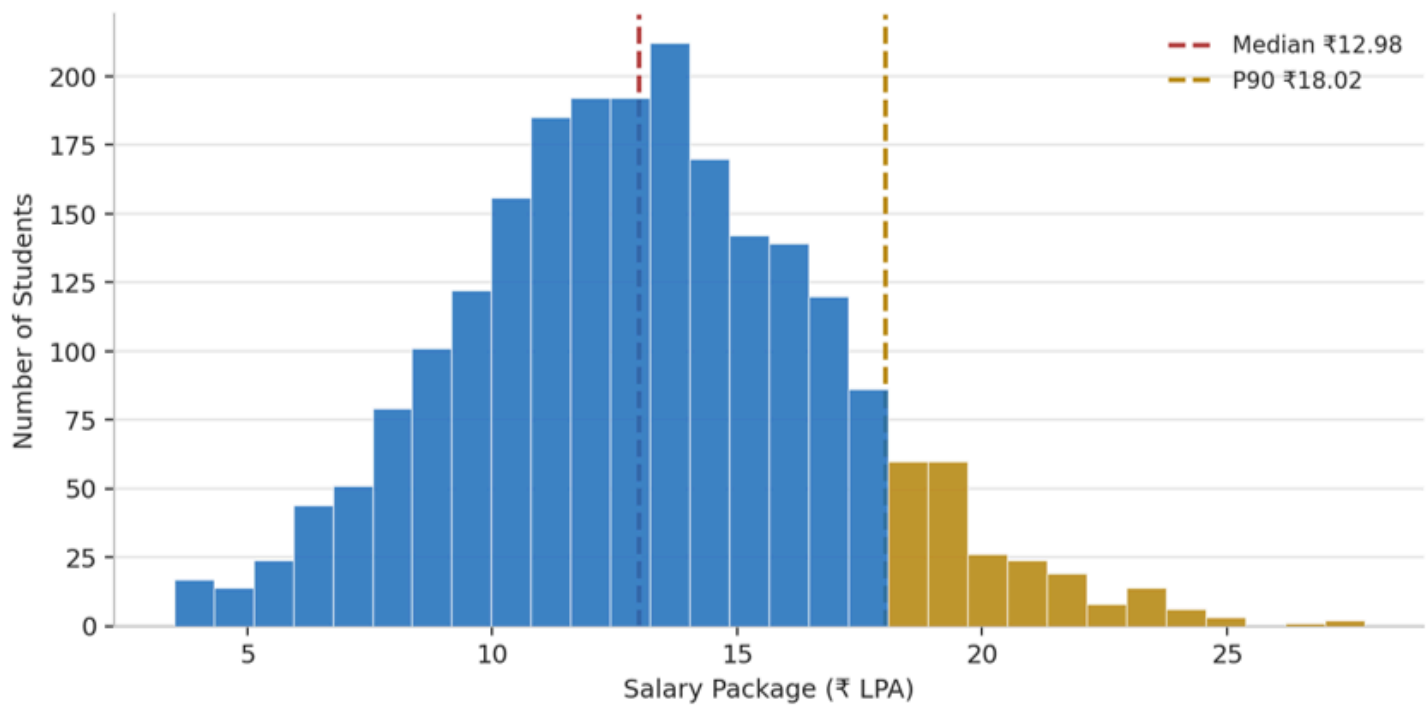
Both metrics show a clear upward trend, but with an interesting inflection point around 3-4 skills: placement rate jumps sharply from about 57.5% (at 2-3 skills) to over 80% at 4 skills, then continues climbing more gradually to nearly 99% at 10 skills. Salary follows a similar but slightly lagged pattern — it dips slightly from 2 to 3 skills, then rises steadily and accelerates toward the end, reaching its steepest climb between 9 and 10 skills (jumping from roughly ₹12.6 LPA to ₹15 LPA).

The overall takeaway is that skill-portfolio breadth matters a great deal, with a critical threshold around 4 skills where placement outcomes improve dramatically, and continued gains in both placement and salary for every additional skill beyond that — with the salary payoff being especially pronounced at the highest skill counts (9-10).

Salary Distribution & Top-Earner Profile

Distribution shape

Statistic	Value (₹ LPA)
Minimum	3.50
10th percentile	8.32
25th percentile (Q1)	10.62
Median (P50)	12.98
Mean	13.16
75th percentile (Q3)	15.66
90th percentile	18.02
95th percentile	19.52
99th percentile	23.02
Maximum	27.80



This histogram shows the overall distribution of salary packages among placed students, with two reference lines marking key benchmarks: the median (₹12.98 LPA, red dashed line) and the 90th percentile (₹18.02 LPA, gold dashed line).

The distribution is roughly bell-shaped but right-skewed, peaking around ₹12-14 LPA where the bulk of students (200+ at the mode) land their packages. Most students fall between ₹8-17 LPA, forming the dense blue-shaded core of the distribution. Beyond the P90 mark of ₹18.02 LPA, shown in gold, the numbers drop off sharply — only a small tail of students secure packages above ₹20 LPA, with very few reaching into the ₹25+ LPA range.

This confirms a typical placement salary pattern: while the median outcome (~₹13 LPA) is solid and represents where most students land, only the top 10% break into significantly higher compensation (₹18 LPA+), with a long, thin tail of high-earners at the very top of the distribution.

The mean (₹13.16 LPA) sits just above the median (₹12.98 LPA), indicating a modest right skew — a smaller number of high earners pull the average up slightly, but the distribution is not dominated by extreme outliers. The interquartile range (Q1–Q3) spans ₹10.62–15.66 LPA, meaning the middle 50% of placed students earn within a roughly ₹5 LPA band.

Profile of the top 10% of earners ($\geq ₹18.02$ LPA, $n = 227$)

Attribute	Top 10% Earners	All Placed Students (for comparison)
Avg. Practical Skill Score	97.4	$\approx 67-74$
Avg. Theoretical Score	76.1	$\approx 60-76$
Avg. Number of Internships	2.47	Dataset-wide average is well under 2
Avg. Number of Certifications	2.25	—
Experience Profile: Both (Hybrid)	90.3%	—
Experience Profile: Internship Only	9.7%	—
Experience Profile: Certification Only / Neither	0.0%	—

Key insight — the top 10% earnings bracket is exclusively hybrid or internship-heavy

Not a single student in the top 10% of earners ($\geq ₹18.02$ LPA) falls into the Certification Only or Neither experience-profile categories. 90.3% are Hybrid and the remaining 9.7% are Internship Only. Combined with a near-maximal average Practical_Skill_Score (97.4 out of 100), this is the clearest single piece of evidence in this report that top-bracket compensation is functionally gated on practical, internship-driven experience — certifications alone cannot reach this tier regardless of how many are stacked.

Recommendations

Based on this analysis, several practical recommendations emerge for different stakeholders. For students, the data suggests treating five months as the practical minimum internship duration when a direct PPO is the goal, since durations shorter than that converted to a direct offer 0% of the time in this dataset. Students should also prioritize breadth of demonstrable skills, aiming for 8–10 skills, while stacking 2–4 certifications, since the skill-count gradient proved to be the single steepest simple predictor found in the analysis. Those attending Tier 2 or Tier 3 institutions with weaker brand recognition should lean especially hard into internships, as the tier-based gap nearly disappears once students have hybrid or internship-only experience profiles. Additionally, when considering smaller or lesser-known internship or certification providers, students should check for ISO certification and MSME/Udyam registration as a quick, verifiable credibility signal.

For the ACC platform and its advisory logic, the recommendation is to extend the AI Pathway Advisor so it explicitly flags the five-month internship-duration threshold when a student's goal is a direct PPO, rather than offering only generic advice to pursue "longer internships." The platform should also add an issuer-credibility checklist — covering ISO and MSME-Udyam status — to the Employer Preference Engine or a dedicated provider-vetting module, particularly for students weighing non-flagship internship or certification options. Furthermore, the certification × internship interaction matrix should be surfaced directly to students choosing between pursuing one more certification versus one more internship, since the data shows these two levers are not equivalent in their impact.

Finally, for placement cells and institutions, the findings point to process efficiency as a controllable lever independent of brand tier — for example, Tier 2 State Tech University showed the slowest average time-to-offer at 56.0 days despite its mid-tier ranking, indicating room for improvement through placement-cell operations alone. Given the near-zero correlation observed between practical and theoretical scores, institutions should also treat internship placement support and certification guidance as two separate, parallel tracks in their curricula and career-services programming, rather than assuming that progress in one area implies progress in the other.

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Conclusion

This analysis reveals several important patterns about what actually drives placement and salary outcomes for students. Institutional tier still carries a real placement and salary premium, but this advantage tracks closely with practical skill score and nearly disappears for students who have hybrid or internship-only experience profiles, suggesting that brand prestige matters far less once practical experience is accounted for. Degree program choice, by contrast, barely moves the needle at all — the spread across five degree types is under ₹0.5 LPA, making it a much weaker lever than tier, experience, or certification-stacking effects. Notably, no statistically significant gender pay gap was found in this dataset ($p = 0.92$), and this parity holds consistently within every experience-profile band, not just in the aggregate, pointing to a genuinely equitable outcome structure rather than one masked by averaging.

Looking at trends over time, outcomes have improved in every graduating year from 2023 to 2026, with 2026 showing a particularly sharp step-change in both practical skill score and offer speed. Certification returns follow a non-linear pattern: the largest placement-probability gains appear from two certifications onward, peaking around four, while salary continues climbing all the way through five. Importantly, certifications and practical skills are statistically independent ($r = 0.08$), which validates a hybrid strategy on quantitative rather than merely anecdotal grounds — a linear regression confirms that both remain strong, non-redundant drivers of salary. Hybrid candidates also receive offers substantially faster, converting 61% quicker on average than certification-only candidates (39 days versus 101 days). Interestingly, internship domain and host-company brand matter far less than internship duration itself, with the six domains and ten major host companies studied clustering tightly together on outcomes, while direct PPOs show a hard duration floor near five months, with no conversions occurring before that point in this dataset.

Among all the factors examined, skill-portfolio breadth — the number of distinct skills listed — emerges as the steepest simple predictor of outcomes in the entire report, more than doubling salary from the low end to the high end. This is reinforced by the composition of top earners: the top 10% bracket is made up entirely of Hybrid or Internship-Only students, with Certification-Only and Neither profiles structurally absent from that tier. Taken together, these findings suggest that internships function as a near-binary gate for strong outcomes — you either have the experiential foundation or you don't — while certifications act more like a continuous multiplier layered on top of whatever internship tier a student has already reached, rather than a substitute for it.